9.4.23

EE25BTECH11011 - Navya

Question:
Find the roots of the quadratic equation graphically.

$$6x^2 - x - 2 = 0 \tag{1}$$

Solution:

$$y = 6x^2 - x - 2 \tag{2}$$

This equation can be represented as the conic:

$$\mathbf{x}^T \mathbf{V} \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0 \tag{3}$$

where

$$\mathbf{V} = \begin{pmatrix} 6 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} -\frac{1}{2} \\ 0 \end{pmatrix}, \quad f = -2 \tag{4}$$

To find the roots, we find the points of intersection of the conic with the x-axis:

$$\mathbf{x} = \mathbf{h} + k_i \mathbf{m} \tag{5}$$

$$\mathbf{h} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{6}$$

Substitute into the conic equation:

$$(\mathbf{h} + k_i \mathbf{m})^\top \mathbf{V} (\mathbf{h} + k_i \mathbf{m}) + 2\mathbf{u}^\top (\mathbf{h} + k_i \mathbf{m}) + f = 0 \tag{7}$$

This simplifies to a quadratic in k_i:

$$k_i^2 \mathbf{m}^T \mathbf{V} \mathbf{m} + 2k_i \mathbf{m}^T (\mathbf{V}\mathbf{h} + \mathbf{u}) + g(\mathbf{h}) = 0 \tag{8}$$

Now substituting the values:

$$k_i = \frac{-\mathbf{m}^T (\mathbf{V}\mathbf{h} + \mathbf{u}) \pm \sqrt{[\mathbf{m}^T (\mathbf{V}\mathbf{h} + \mathbf{u})]^2 - \mathbf{m}^T \mathbf{V} \mathbf{m} \cdot g(\mathbf{h})}}{\mathbf{m}^T \mathbf{V} \mathbf{m}} \tag{9}$$

$$\Rightarrow k_i = \frac{-(-\frac{1}{2}) \pm \sqrt{(-\frac{1}{2})^2 - 6(-2)}}{6} = \frac{\frac{1}{2} \pm \sqrt{\frac{1}{4} + 12}}{6} = \frac{\frac{1}{2} \pm \frac{7}{2}}{6} \tag{10}$$

$$\Rightarrow k_1 = \frac{4}{6} = \frac{2}{3}, \quad k_2 = \frac{-3}{6} = -\frac{1}{2} \tag{11}$$

Hence, the real roots are:

$$\mathbf{x}_1 = \begin{pmatrix} \frac{2}{3} \\ 0 \end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix} -\frac{1}{2} \\ 0 \end{pmatrix} \tag{12}$$